An Event-driven and User Behavior-empowered Hybrid Recommendation System Based on Voluntary Service

Mingqian Yue[1], Xing Zhang[1,*], Jiaxin Zhang[1] and Junbo Zhang[2]
[1]*Key Laboratory of Universal Wireless Communication, School of Information and Communication Engineering, Ministry of Education, Beijing University of Posts and Telecommunications, Beijing 100876, China*
[2]*JD Intelligent Cities Research, China and JD Intelligent Cities Business Unit, JD Digits, China*
**E-mail: zhangx@ieee.org*

***Abstract*—Despite the popularity of voluntary service, the voluntary recommendation domain is less explored. Besides, compared with pop recommendation fields, the dataset from the Chinese voluntary service information system shows specific features of voluntary service, summarised as instability of volunteers and time-effectiveness of volunteer projects. These features lead to problems such as data sparsity and cold start, which makes it difficult for traditional recommendation algorithms to achieve good results, while regular hybrid systems also face the problem of scalability with the data accumulating. Based on such findings, we propose a layered multi-graphs structure and construct a novel hybrid recommendation system. The proposed system helps to alleviate the data sparsity problem and improve algorithm execution efficiency through graph-based methods such as community detection, and solve the cold start problem and enhance the recommendation performance by combining the advantages of proper algorithms. By simulating the actual daily activities of volunteers in our experiments, it is shown that the recommendation success rate of the proposed system reaches 53% with a better ability to deal with massive data.**

***Index Terms*—voluntary service; layered multi-graphs structure; hybrid recommendation system;**

I. Introduction

Since Beijing hosted the Olympic Games in 2008, people have become more aware of the importance of volunteering [1]. Over the past decade, approximately **120 million** people registered to be volunteers, **2.3 million** volunteer projects were successfully launched, and there exists accumulated **80.3 million** interactive records of them. There is no doubt that the volunteers desperately need a recommendation system to help them find the ideal choice from the vast amount of projects. However, from the best of our knowledge, there have been few detailed investigations of voluntary service recommendations. The only relevant study we could find is [2], which employed the PersonalRank [3] algorithm with basic attributes of volunteers and projects to make recommendations. But the extracted dataset it explored is small and the discussion of the inherent characteristics of the voluntary service field is not enough.

Under such a background, we firstly analyze the dataset from the Chinese voluntary service information system to summarize the basic features and problems of this area. Then, we propose a layered multi-graphs structure and construct the event-driven and user behavior-empowered hybrid recommendation system(EURS) to deal with the problems from the level of the system architecture. Furthermore, our experiments show that compared with regular hybrid systems, our system performs better with higher efficiency.

The rest of the paper is organized as follows. Section 2 briefly reviews the related work. Section 3 describes the voluntary service dataset and summarizes the characteristics and problems of this area. In Section 4, we introduce the details of our system structure and the adjustments of different algorithms. Evaluation results and discussions are presented in Section 5 and we draw our conclusion in the last section.

II. Related work

Collaborative Filtering is widely adopted by a variety of applications since it's simple and effective. It mainly includes two categories: memory-based and model-based methods [4]. Memory-based methods utilize the heuristic-based search strategy to find users with similar tastes to the current user [5] or the similar items that the user might be interested in, while model-based methods focus on learning an off-line model using different machine learning techniques such as clustering or matrix factorization. With its simplicity, it faces the problem of data sparsity and cold start when there's not enough interaction data of users and items. One way to overcome such shortcomings is to exploit other recommendation methods such as content-based methods [6] or merge multi-source heterogeneous data such as extra demographic information [7]. Graph-based models are another proper choice, which turns the recommendation task into the node selection task on heterogeneous graphs or homogeneous graphs [8]. The Random Walk [3] employed in [2] is one of the most widely used related algorithms [9]. Moreover, hybrid recommendation systems are always ideal solutions to improve the quality of the recommendations [10] by combining the advantages of different algorithms. Inspired by these studies and our analysis of voluntary service, we put forward a layered multi-graphs structure and build the practical recommendation hybrid system EURS to address existing problems of voluntary service.

III. Data introduction and analysis

The volunteerism data we explore for the study, which is collected from 2008 to 2018, comes from the Chinese voluntary service information system, and mainly includes information about volunteers and projects. Some of the information is structured: such as volunteers' province, district, gender, birthdate, groups, education background, and student flag; projects' province, district, organizer groups, recruit start/end date, and so forth. Some are unstructured such as projects' overviews and target people of service. Through analyzing these data, the following two main features have been found.

A. Instability of Volunteers

Through our analysis, over the past ten years, about **67.56%** of volunteers participated in fewer than 3 projects, which indicates that unlike the other recommendation fields, where people may buy multiple products, or like multiple movies and music, volunteers normally will not participate in many projects in their lifetime. And we randomly took $10,000$ volunteers to count their time interval of participating projects and found that about **64.05%** of volunteers' time interval of participating projects are in three months, suggests that many volunteers' interests in volunteering only last for a short period.

From another point of view, about an average of **96,000** newly registered volunteers joined the voluntary service system every day in 2018. For these cold start volunteers, many recommendation algorithms can not work due to their lack of historical interactive records. These findings reveal the serious data sparsity and cold start problems of voluntary service.

B. Time-effectiveness of Volunteer Projects

Across the country, there are thousands of projects' dynamic changes in establishing or finishing every day, such as from 2016 to 2018, the average daily project ending of about **1,820**. Correspondingly, the average project durations(online duration represents the number of days a project lasts from start to finish) in 2018 is **43** days, with which less than or equal to 7 days, that is, one-week accounts for **52.2%** of the total number of projects, which proves that the volunteer projects are mostly short-term projects. These results indicate the dynamic changes of projects in this field. More than that, such a dynamic feature is rarely seen in other areas of recommendation. Old music or books can still be recommended to users, while there is no point in recommending finished projects to volunteers, which is a kind of temporal characterization with invalidation significance. We refer to this project feature as dynamic time-effectiveness.

IV. System description

The instability of volunteers and the time-effectiveness of projects invalidate many models that reflect long-term stable interests of users, while most algorithms that reflect users' current interests can hardly achieve good results. Moreover, regular hybrid systems will greatly improve the algorithm's time consumption. Therefore, we propose the layered multi-graphs structure to deal with these problems.

A. Layered Multi-graphs Structure

We follow the following steps to construct the layered multi-graphs structure(LMGS) as shown in Fig. 1.

1) We convert the interaction records of the main database to the overall graph(OvG) and build the online graph(OnG) on top of OvG to store the relevant interaction records of the online projects and active volunteers. The nodes of the graph represent the entities of volunteers and projects, and the links represent the participation relationship between them.

2) We maintain the online project attribute graph(OPAG) and the online volunteer attribute graph(OVAG) on top of OnG. In the construction process, we objectify the attributes of volunteers and projects. For structured attributes, after cleaning and simply transforming, such as converting the volunteers' birthdate to age, we directly map them to the graph as entities. For unstructured attributes, we process them by word cutting and stop-word filtering to get the word segmentation list and then map each word to the graph as entities. For example, pupils and the elderly are processed words from the attribute, projects' target people of service. It is clearly more appropriate to think of them as entities rather than just semantic words.

3) These graphs are stored in the Neo4j database and we realize real-time update of these databases: if the projects of online databases reach the deadline, the corresponding records will be truncated from the related databases; if a new project/volunteer is registered, the relevant information will be added to the related databases.

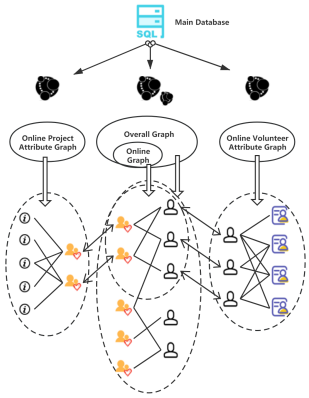


Fig. 1. Layered Multi-graphs Structure.

B. Hybrid Recommendation System

Considering there have been few relevant studies in the field of voluntary service [2] and many state-of-art models may not be necessarily better than traditional models in many scenarios declared in [11], we compared some traditional models and ultimately adapt PersonalRank(PR) [3], ContentBased-CF(CBCF) [6], UserCF(UCF) [5] and DemographicBased-CF(DBCF) [7] to construct the recommendation system at the present stage. The implementation of these algorithms is based on LMGS.

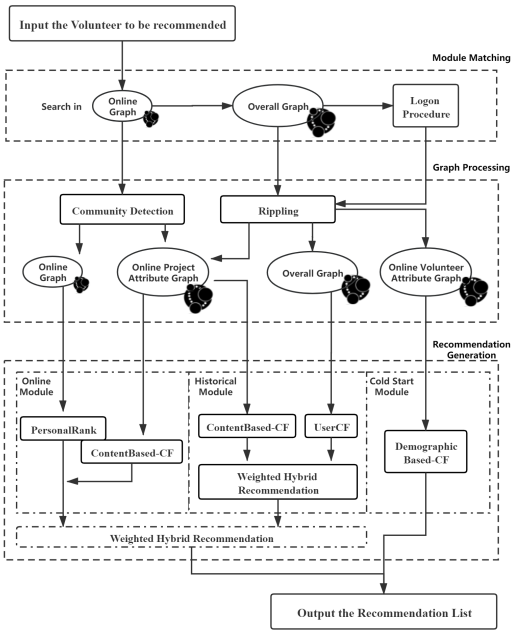


Fig. 2. Framework of Hybrid System.

The proposed system mainly consists of three stages and three modules as shown in Fig. 2, which is a series process. When a volunteer login into the system, the system will search the volunteer in OnG first: if there exist relevant records, corresponding graph processing steps and the online module will execute and then turn to the OvG matching; otherwise, the system will directly search for the volunteer in OvG to determine to execute the historical recommendation steps or cold start steps.

In the process, the "Module Matching" stage determines which recommendation module ultimately generates the recommendation list, and "Graph Processing"(GP) supports the recommendation module to improve execution efficiency. Actually, the "GP" stage is also part of the corresponding recommendation module and the details of the implementation of them will be talked together in the following parts. Table I lists the main mathematical symbols we use.

TABLE I
DEFINITIONS OF MATHEMATICAL SYMBOLS

Symbol	Definition
V	The set of all volunteers. $\forall v_i, v_j \in V$
P	The set of all volunteer projects. $\forall p_i, p_j \in P$
$P(v_i)$	The set of projects that volunteer v_i has participated in. $Pv_i \subset P$
$Atr(v_i), Atr(p_j)$	The attributes set of volunteer v_i or project p_j respectively. $\forall attr_i, attr_j \in Atr$
$EoA_{attr_i}(.)$	The processed entity set of attribute $attr_i$ of v_i or p_j.

1) Online Module: This module recommends projects for active volunteers currently participating in online projects. We refer to [2] to choose PR as the core algorithm for this module. However, PR has its drawbacks: in general, the frequency of the random walk is fixed, the too-small frequency will affect the recommendation accuracy, and too-high frequency will increase the cost time. Aiming at this problem, we introduced Louvain algorithm [12] to divide the interactive network into communities, which has been discovered to perform well in [13], and add the limitation $frequency \propto SC(v_i)$. We call such combination as PRBL(PersonalRank Based on Louvain community), where $SC(v_i)$ represents the size of the community to which the volunteer v_i belongs and here we define the size of the community as the number of projects in the community. Dynamically walking randomly according to the community size helps to improve the implementation efficiency on the premise of ensuring the recommendation accuracy.

But PRBL faces another problem: some interactive subgraphs exist in isolation, which causes some lists of recommendations to be inadequate. Therefore, we also adapt the Louvain detection on OPAG to find the project-attribute communities to get their similarities of each other by (1) as shown in Fig. 3 (a), where the subscript c_k represents the community to which the entity belongs. And then we can complement the recommendation list by Collaborative Filtering.

$$sim\left(p_{i,c_i}, p_{j,c_j}\right) = \begin{cases} \sum\limits_{attr}^{Atr(p_{i,c_i}) \bigcap Atr(p_{j,c_j})} sim_{attr}\left(p_{i,c_i}, p_{j,c_j}\right), \\ \qquad\qquad (c_i = c_j) \\ 0, Otherwise \end{cases} \tag{1}$$

$$sim_{attr}\left(p_{i,c_i}, p_{j,c_j}\right) = \frac{\left|EoA_{attr}\left(p_{i,c_i}\right) \bigcap EoA_{attr}\left(p_{j,c_j}\right)\right|}{\sqrt{\left|EoA_{attr}\left(p_{i,c_i}\right)\right| \left|EoA_{attr}\left(p_{j,c_j}\right)\right|}} \tag{2}$$

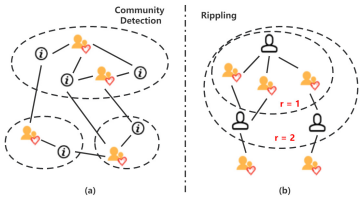


Fig. 3. An illustration of neighbors finding.

2) Historical Module: This module recommends projects for those volunteers who have volunteered before. For such a recommendation scenario, we apply UCF and CBCF since they can reflect volunteer relationships or interests simply through historical information and do not require current interactive records.

Considering the data sparsity of this field, we first get the volunteer group that has IC intersecting connections with the current volunteer by rippling two laps on OvG as shown in Fig. 3 (b) ($r = 2$), and then get the similarities between them by $sim\left(v_i, v_j\right) = \frac{|P(v_i) \cap P(v_j)|}{\sqrt{|P(v_i)||P(v_j)|}}$. Where IC is a parameter to

control the filtering degree, and we set it as 1 here, for OvG is a relatively sparse graph. Such implementation alleviates data sparsity to a certain extent and improves the efficiency of UCF.

For CBCF, similar to the above operation, we find online project communities that have attributes intersections with the current project by "rippling" ($IC = 1$) on OPAG, and then calculate the similarity between the projects by (1). Here p_{i,c_i} represents the finished projects, the subscript $c_i = c_a \cup c_b \cup ...$ represents those communities that have two lap connections with p_{i,c_i}, and p_{j,c_j} represents the online project that belongs to c_j.

Finally, we combine UCF and CBCF to generate the recommendation list as (3), where $N(v_i)$ represents the nearest neighbors of v_i.

$$\begin{aligned} rec(v_i, p_j) = \beta * \sum_{v_k \in N(v_i)} \sum_{p_j \in P(v_k)} sim(v_i, v_k) \\ +(1-\beta) * \sum_{p_{i,c_i} \in P(v_i)} sim\left(p_{i,c_i}, p_{j,c_j}\right) \end{aligned} \tag{3}$$

3) Cold Start Module: This module recommends projects for cold start volunteers who have never participated in any projects. In many studies, researchers often provide hot recommendations, but it is non-personalized and its performance is always limited. In our experiments, we adapt DBCF to solve such a problem. For newly registered volunteers, we will ask them to finish the logon procedure to get their basic information, such as age, gender, district, etc. And we also apply the "rippling" mentioned above on OVAG to get the volunteer group that has intersecting attributes with the current volunteer. Here we set $IC = 5$. Then, we can further screen out the most similar active volunteers through (4) and make recommendations.

$$sim(v_i, v_j) = \frac{\left|(\cup_{attr}^{Atr(v_i)} EoA_{attr}(v_i)) \cap (\cup_{attr}^{Atr(v_j)} EoA_{attr}(v_j))\right|}{\sqrt{\left|(\cup_{attr}^{Atr(v_i)} EoA_{attr}(v_i))\right| \left|(\cup_{attr}^{Atr(v_j)} EoA_{attr}(v_j))\right|}} \tag{4}$$

V. EXPERIMENTS AND RESULT ANALYSIS

To test and evaluate our proposed system, we used the real data provided by the Chinese voluntary service information system.

In the experiments, interactive records of volunteers in Beijing are extracted as the experimental dataset, which takes into account the regional nature of voluntary service. As the analysis shows that the projects with a duration of fewer than 7 days account for **52.2%**, we set one week as the simulation time and extract the data of the first three weeks as history records. We set the simulation week as the last week of May 2018, that is, May 21-27, to avoid the influence of statutory holidays and special periods such as winter or summer holidays. The training/test dataset is extracted and divided randomly in days during the simulation week. The main point of our test method is the test set of someday will be added as the training set of the following days, and the completed projects will be removed from the training set timely to simulate the daily changes of the real world.

Table II describes the specific data extraction results. As can be seen from the table, the number of the training set and volunteers and the proportion of cold-start volunteers in the daily test set confirms the serious data sparsity and cold start problems in this field respectively as previous analysis.

TABLE II
EXTRACTED DATASET

Date	5.21	5.22	5.23	5.24	5.25	5.26	5.27
Volunteers	95,128	96,768	98,629	100,422	102,539	104,394	105,750
Projects	3746	3813	3892	3954	4035	4092	4131
Training Set	125,919	129,114	132,134	135,376	138,625	142,217	145,577
Test Set	3195	3020	3242	3249	3592	3360	2689
Cold-Start Volunteers	58.2%	57.7%	57.8%	57.0%	58.0%	61.2%	59.1%

A. Test Methods and Indicators

Since participation relationship belongs to implicit feedback, we adapt Top-N test indicators in the experiments such as success_rate (5), recall (6), coverage (7), and popularity (8). The success_rate is a conversion form of accuracy. Based on the actual situation, the average number of projects that volunteers participate in every day in 2018 is **1.61**, and the proportion of fewer than 3 is as high as **93.1%**. In test dataset, the corresponding values are **1.38** and **92.34%**. In such a scenario, we believe that as long as we successfully recommend one project to a volunteer which he is truly interested in, that is a successful recommendation. Hence, the recommended project number N is set as **3**. And we also set time-consuming(TimeC) as one important indicator, which represents the average time the system costs to generate recommendations for each volunteer.

We did the experiments and got the index results in days, and finally got the average value to avoid the contingency brought by the particularity of data extraction.

$$Success = \frac{\sum_{v \in V} \begin{cases} 1, & |R(v) \bigcap T(v)| > 0 \\ 0, & |R(v) \bigcap T(v)| = 0 \end{cases}}{|V|} \tag{5}$$

$$Recall = \frac{\sum_{v \in V} |R(v) \bigcap T(v)|}{\sum_{v \in V} |T(v)|} \tag{6}$$

$$Coverage = \frac{|\cup_{v \in V} R(v)|}{|P|} \tag{7}$$

$$Popularity = \frac{\sum_{v \in V} \sum_{p \in R(v)} \log(1 + Hit(p))}{\sum_{v \in V} |R(v)|} \tag{8}$$

Where $R(v)$ is the list of recommended projects generated by the recommendation system for the volunteer v, and $T(v)$ is the list of projects participated by the volunteer v recorded in the test set. V stands for the test set of volunteers and P for online volunteer projects that are recruiting volunteers. $Hit()$ means the current number of participants in the volunteer project.

TABLE III
EURS vs. TRADITIONAL SYSTEMS

	Success	Recall	Coverage	Popularity	TimeC(s)
MostPop	14.71%	14.21%	2.05%	7.021	0.0003
PR	27.97%	27.47%	51.25%	3.995	1.7803
CBCF	24.32%	23.90%	65.39%	3.572	0.1184
UCF	19.54%	19.40%	45.72%	3.479	0.1210
DBCF	36.59%	35.46%	68.63%	5.139	2.9759
PD	44.30%	43.03%	69.78%	5.118	4.7562
UCD	52.74%	51.53%	**83.53%**	**4.687**	3.2154
UPD	53.30%	52.03%	77.46%	4.958	4.8772
EURS	**53.47%**	**52.18%**	75.50%	4.725	**0.3784**

B. Experimental Results

In the experiments, each regular algorithms and hybrid systems are implemented after filter processing according to the time-effectiveness of projects. We set the algorithm parameters according to the real data analysis results. The nearest neighbor value for UCF is set as **50**, for we have counted that about an average of 50 volunteers participate in one project. And we set the nearest neighbor value for CBCF as **10**, since we have known that volunteers will not participate in many projects at the same time.

As shown in Table III, it is obvious that any recommendation algorithm performs better than hot recommendation(MostPop), and the hybrid systems are always superior to the separate traditional algorithm, where PD represents the weighted hybrid system combining with PR and DBCF, UCD represents the weighted hybrid system combining with UCF, CBCF, and DBCF and UPD is the same thing. Hybrid systems all adapt DBCF to solve the cold start problem and combine the advantages of different algorithms to improve performance from different aspects, for UCF is from the volunteers' perspective and CBCF is from the projects' perspective. And as expected, hybrid systems greatly improve the recommendation time consumption, which is intolerable for online web service.

For the two main indicators, success rate and recall rate, EURS outperforms all the other hybrid systems. More importantly, it consumes just a tenth as much time as the other hybrid recommendation systems and even better than some separate algorithms such as PR, which represents stronger scalability brought by the graph's structure. However, in terms of coverage and popularity, EURS performs mediocre, which may be related to the fact that we focus the algorithms on small scale graphs to improve the execution efficiency. Besides, LMGS could also help to show the reasons for recommending projects to each volunteer in a visual form, with stronger explanatory ability.

Actually, we also applied model-based algorithms such as pureSVD and BPR to make recommendations. As expected, such kinds of algorithms are more susceptible to data sparsity, with the recommendation success rate of less than 1%, similar to the random recommendation. Therefore, at this stage, we do not discuss them.

VI. CONCLUSIONS

In recent years, the concept of Urban Computing has attracted more attention. Researchers try to realize the intelligence of the city by studying the data from different fields. Voluntary service is such an area that has not got much attention but has the potential to influence Urban Computing because it's an important part of social life. In this paper, we summarize the specific characteristics of voluntary service, construct a layered multi-graphs structure and propose the event-driven and user behavior-empowered hybrid recommendation system, which helps to solve the problems of data sparsity and cold start with strong scalability. The framework is of practical value and we believe it provides a reference in other recommendation fields with similar characteristics.

ACKNOWLEDGMENT

This work is supported by the National Science Foundation of China under Grant 61771065.

REFERENCES

[1] https://www.chinavolunteer.cn/.
[2] F. Tian, Y. Chen, X. Wang, T. Lan, Q. Zheng, and K.-M. Chao, "Common features based volunteer and voluntary activity recommendation algorithm," in *2015 IEEE 12th International Conference on e-Business Engineering*. IEEE, 2015, pp. 43–47.
[3] D.-C. Nie, Y. Fu, J.-L. Zhou, Z. Liu, Z.-K. Zhang, and C. Liu, "A personalized recommendation algorithm via biased random walk," in *2014 11th International Joint Conference on Computer Science and Software Engineering (JCSSE)*. IEEE, 2014, pp. 292–296.
[4] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: a survey of the state-of-the-art and possible extensions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734–749, 2005.
[5] Z.-D. Zhao and M.-S. Shang, "User-based collaborative-filtering recommendation algorithms on hadoop," in *2010 Third International Conference on Knowledge Discovery and Data Mining*. IEEE, 2010, pp. 478–481.
[6] P. Lops, M. De Gemmis, and G. Semeraro, "Content-based recommender systems: State of the art and trends," in *Recommender systems handbook*. Springer, 2011, pp. 73–105.
[7] X. W. Zhao, Y. Guo, Y. He, H. Jiang, Y. Wu, and X. Li, "We know what you want to buy: a demographic-based system for product recommendation on microblogs," in *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2014, pp. 1935–1944.
[8] S. Shaikh, S. Rathi, and P. Janrao, "Recommendation system in e-commerce websites: A graph based approached," in *2017 IEEE 7th International Advance Computing Conference (IACC)*. IEEE, 2017, pp. 931–934.
[9] C. Eksombatchai, P. Jindal, J. Z. Liu, Y. Liu, R. Sharma, C. Sugnet, M. Ulrich, and J. Leskovec, "Pixie: A system for recommending 3+ billion items to 200+ million users in real-time," in *Proceedings of the 2018 world wide web conference*, 2018, pp. 1775–1784.
[10] W. Shalaby, B. AlAila, M. Korayem, L. Pournajaf, K. AlJadda, S. Quinn, and W. Zadrozny, "Help me find a job: A graph-based approach for job recommendation at scale," in *2017 IEEE International Conference on Big Data (Big Data)*. IEEE, 2017, pp. 1544–1553.
[11] M. F. Dacrema, P. Cremonesi, and D. Jannach, "Are we really making much progress? a worrying analysis of recent neural recommendation approaches," in *Proceedings of the 13th ACM Conference on Recommender Systems*, 2019, pp. 101–109.
[12] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of statistical mechanics: theory and experiment*, vol. 2008, no. 10, p. P10008, 2008.
[13] M. Fatemi and L. Tokarchuk, "An empirical study on imdb and its communities based on the network of co-reviewers," in *Proceedings of the First Workshop on Measurement, Privacy, and Mobility*, 2012, pp. 1–6.